Q1

Fill 3G, pour to 4G. Fill 3G again, pour until 4G full leaving 2G in 3G. Empty 4G. Pour remaining 2G into 4G. Result: 4G jug contains exactly 2 gallons.

Q2

i) Percepts: camera images, terrain, temperature, chemical sensors, position, battery.
ii) Environment: partially observable, sequential, dynamic, uncertain, real-world.
iii) Actions: move, drill, collect samples, analyze, capture images, transmit data.
iv) Performance: safe navigation, successful sample collection, accurate analysis, energy efficiency.
v) Model-based goal-based agent because it maintains an internal model and plans actions.

Q3

State: arrangement of 8 queens. Initial state: empty board. Actions: place one queen in next column. Goal: 8 queens with no row/column/diagonal attacks. Path cost: one per placement or constant. Search: Backtracking/DFS.

Q4

Agent: Goal-based problem-solving agent. Pseudocode: Input source,destination,cab types; get available cabs; evaluate fare,time,user preference; choose best cab; confirm booking; display details.

Q5

Uniform Cost Search expands the node with the lowest cumulative path cost first until destination G is reached. Maintain a priority queue ordered by path cost, expand lowest-cost node, update costs of neighbors, stop when G is removed from the queue. The exact least-cost path depends on the weighted graph provided.